ECO375 Term Project — Alternative Model Spec

Two-way fixed effects: ERP removed, week fixed effects added

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Companion to the main tracker (`eco375_tracker.tex`). Tags: [data] = computed from our files; [judgment] = a reasoned choice.

1 Specification

Stock i, week t (weeks end Friday):

$$Y_{it} = \beta_1 \,\mathrm{AbnAttn}_{it} + \beta_3 \,\mathrm{Size}_{it} + \beta_4 \,\mathrm{BM}_{it} + \beta_5 \,\mathrm{Turnover}_{it} + \beta_6 \,\mathrm{InstHold}_{it} + \beta_7 \,\mathrm{NumEst}_{it} + \beta_8 \,\mathrm{SUE}_{it} + \beta_9 \,\mathrm{Attn}_{i,t-1} + \beta_{10} \,\mathrm{Dvol}_{it} + a_i + \lambda_t + u_{it}$$

- a_i = firm fixed effect: anything about stock i that doesn't change over the sample.
- λ_t = week fixed effect: anything that hits *every* stock equally in week t.
- ERP_t is dropped; β_2 is retired, and the other coefficients keep their numbers from the main spec.
- Y_{it} can be the raw weekly return `ret_w`. Subtracting r_f changes nothing: $r_{f,t}$ is the same for every stock in week t, so λ_t absorbs it, and β_1 comes out identical with or without it.

Reading of β_1. In a week when stock i has an attention spike, its return is β_1 percentage points higher than (a) other stocks' returns that same week and (b) its own usual return, holding the controls fixed. Each stock is compared both with itself (via a_i) and with the market that week (via λ_t).

2 What the week fixed effect does

ERP is one number per week shared by all stocks, so it is just one of the things λ_t soaks up. Including both is impossible: ERP would be perfectly collinear with the week dummies and drop out. λ_t also absorbs:

- the market return, r_f, and any macro shock (Fed meetings, CPI releases, trade-war headlines);
- weeks when news is heavy for *all* stocks at once, such as earnings season or market-wide selloffs;

- changes in how hard the scraper collected news, *to the extent* they hit all tickers equally. [data] Total ticker-tagged articles fall from 580k (2019) to 169k (2021), and λ_t absorbs the common part of that swing. Ticker-specific gaps (e.g. no AAPL news before 2020) are *not* absorbed. They are still handled by the `covered` and `in_gap` filters.

What it does not do. λ_t is still one number per week shared by all stocks, so it assumes every stock reacts to a common shock by the same amount. A high-beta stock's extra market move, $(\beta_i^{\text{mkt}} - \bar{\beta}^{\text{mkt}}) \times \text{market}_t$, stays in u_{it}. If high-beta stocks also get more coverage in big market weeks, β_1 picks some of that up. [judgment] Section 5 gives two fixes.

3 What variation is left to estimate β_1

After removing a_i and λ_t, β_1 is estimated only from how a stock's attention in a given week *differs* from (its own normal level) and (what is typical across stocks that week). Two consequences:

- Any regressor that is the same for all stocks in a week drops out automatically. That covers ERP, r_f, VIX, interest rates and market-wide sentiment.
- Slow-moving controls (BM, InstHold, NumEst, to a lesser degree Size) have little variation left once a_i is removed. Their coefficients will be noisy. That's acceptable, since they're there to protect β_1, not to be interpreted.

4 Estimation

- **Method:** OLS with firm and week effects absorbed. The panel is unbalanced (stocks enter and leave coverage), so simple double-demeaning is not exact. Use software that absorbs both sets of effects properly:
  - Python: `linearmodels.PanelOLS(..., entity_effects=True, time_effects=True)`
  - Stata: `reghdfe y x1 x2 ..., absorb(firm week) vce(cluster firm)`

  With about 104 weeks, adding week dummies directly (`C(week)`) alongside firm demeaning also works.
- **Standard errors:** cluster by firm, since a stock's errors are correlated over time. Two-way clustering by firm and week is a robustness check. λ_t removes the average common shock, but not shocks shared within industries.
- **Sample:** 2018–2019 ($T = 104$), covered weeks only, outside scraper gaps, no broken prices (`covered`= 1, `in_gap`= 0, `bad_px`= 0).
- **Returns:** winsorize `ret_w` at the 1st/99th percentile. [data] The raw panel has weekly returns above +100% from bad split adjustments.

5 Variants for unequal market sensitivity

	Change to the spec	What it fixes / costs
A. Baseline	As above: $a_i + \lambda_t$, raw Y	Simplest. Assumes all stocks react equally to common shocks.
B. Abnormal return	$Y_{it} = r_{it} - r_{f,t} - (\hat{b}_i \, \text{Mkt}_t + \hat{s}_i \, \text{SMB}_t + \hat{h}_i \, \text{HML}_t)$, loadings estimated per stock on 2017 weekly data; keep $a_i + \lambda_t$	Each stock gets its own market, size and value sensitivity. Costs: needs a clean 2017 history per stock (drops new listings), and assumes 2017 loadings still hold in 2018–19.
C. Stock-specific market slope	Add $\gamma_i \cdot \text{Mkt}_t$ (market return interacted with firm dummies); keep $a_i + \lambda_t$	Sensitivities estimated in-sample, so no reliance on old data. Costs: one extra coefficient per stock (fine with 104 weeks), and the market term absorbs some variation.

[judgment] Report A as the main result and B as the robustness check. If $\hat{\beta}_1$ barely moves between them, unequal betas aren't driving the result.

6 Data status for this spec

Term	Panel column	Status
Y	`ret_w`	**Built**
AbnAttn	`abn_attn`	**Built**
Attn_{t-1}	`n_news_lag1`	**Built**
Dvol	`dvol_w` (use log)	**Built**
λ_t, a_i	`week_end`, `ticker`	**Built**
Size, BM, Turnover, InstHold, NumEst, SUE	—	**Download** (deferred)
Variant B factors	Ken French weekly Mkt, SMB, HML, RF	**Download**

A first run is possible today with Y, AbnAttn, lagged attention, Dvol and both fixed effects. The remaining controls are added as they arrive.

7 Notes carried over from earlier discussion

- Volume controls (Turnover, Dvol) measured in the *same* week partly reflect the attention itself. Use last week's values. [judgment]
- Same-week news and returns overlap. Run with all articles and with price-report headlines removed (`headline_classifier.py`), plus next week's return as a timing check. [data] Price reports are 15% of 2018–19 articles; 21% follow a day with a move $> 3\times$ normal, vs. 11% for headlines with no price mention.